

324083245 | see pdf

# I M L - theoretical sec

1.1: 1)  $f_1, \dots, f_m: C \rightarrow \mathbb{R}$  convex  $\gamma_1, \dots, \gamma_m \in \mathbb{R}_+$

$$\text{קמור} \quad \sum_{i=1}^m \gamma_i f_i(x) = g(x) \quad (3)$$

$x, y \in C, t \in [0, 1]$  ו:

$$\begin{aligned} g(tx + (1-t)y) &= \sum_{i=1}^m \gamma_i f_i(tx + (1-t)y) \leq \sum_{i=1}^m \gamma_i (tf_i(x) + (1-t)f_i(y)) \\ &= \sum_{i=1}^m \gamma_i t f_i(x) + \sum_{i=1}^m \gamma_i (1-t) f_i(y) = t \sum_{i=1}^m \gamma_i f_i(x) + (1-t) \sum_{i=1}^m \gamma_i f_i(y) \\ &= t g(x) + (1-t) g(y) \quad \square \end{aligned}$$

2) קמור  $h(x) = -(\lambda^2) = -x^2$  קמור  $f(x) = -x$  קמור  $g(x) = x^2$

1.2:

קמור  $\max(0, 1 - \gamma(x^T W + b))$  (3)

ניגון קמור  $1 - \gamma(x^T W + b)$  פונקציה קמורה פונקציה קמורה (הוכחה)

קמור  $\max(f, g)$  קמור  $f, g$  קמור  $\max(0, 1 - \gamma(x^T W + b))$

0 פונקציה קמורה קמור  
מתקן למקרים.

$f = 1 - \gamma(x^T W + b), f(w, b) < 0$  כולל

$$\begin{aligned} f &= 1 - \gamma x^T W - \gamma b, \quad \frac{\partial f}{\partial b} = -\gamma \\ &\quad \frac{\partial f}{\partial W} = -\gamma x^T \end{aligned} \Rightarrow \begin{pmatrix} -\gamma \\ -\gamma x^T \end{pmatrix} = \nabla f(w, b)$$

$$g = \begin{cases} \begin{pmatrix} 0 \\ -\gamma x^T \end{pmatrix} & f(w, b) \leq 0 \\ \begin{pmatrix} -\gamma \\ -\gamma x^T \end{pmatrix} & f(w, b) > 0 \end{cases} \Leftrightarrow \nabla f(w, b) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \begin{matrix} f(w, b) < 0 \text{ כולל} \\ f(w, b) = 0 \text{ כולל} \end{matrix}$$

$$\forall k \in [m] \quad g_k \in \partial f_k(x) \quad \text{כאשר } f_1, \dots, f_m: \mathbb{R}^d \rightarrow \mathbb{R} \quad (5)$$

$$\forall x \in \mathbb{R}^d \quad \sum_{k=1}^m g_k \in \partial \sum_{k=1}^m f_k(x) \quad \text{כי } f(x) = \sum_{k=1}^m f_k(x)$$

$$f_k(y) \geq f_k(x) + g_k^T(y-x) \quad : (1) \text{ כי } x \rightarrow y_0 \text{ כך } y_0 < x \quad \text{אז}$$

$$\sum_{k \in [m]} f_k(y) \geq \sum_{k \in [m]} (f_k(x) + g_k^T(y-x)) = \sum_{k \in [m]} f_k(x) + \sum_{k \in [m]} g_k^T(y-x)$$

$$\therefore \partial \sum_{k \in [m]} f_k(x) \ni \sum_{k \in [m]} g_k, \quad (1) \text{ כי } x \rightarrow y_0 \text{ כך } y_0 < x \quad \text{אז}$$

$$f(w, b): \mathbb{R}^d \rightarrow \mathbb{R} = \frac{1}{m} \sum_{i=1}^m \ell_{x_i, y_i}^{h, \text{net}}(w, b) \quad \text{כאשר } S = \{(x_i, y_i)\}_{i=1}^m \subseteq \mathbb{R}^d \times \{\pm 1\} \quad (6)$$

$$\partial_w \ell(w, b) = \frac{1}{m} \sum_{i=1}^m \left( \begin{cases} (-y_i; x_i^T) & \text{if } \ell_{x_i, y_i}^{h, \text{net}}(w, b) > 0 \\ (0) & \text{if } \ell_{x_i, y_i}^{h, \text{net}}(w, b) < 0 \end{cases} \right) \quad \text{אז } \partial_w$$

$$\text{ReLU activation} = \sigma \quad b \in \mathbb{R}^m \quad w \in \mathbb{R}^{m \times d} \quad h = \sigma(w x + b) \quad x \in \mathbb{R}^d \quad (7)$$

$$c \in \mathbb{R} \quad u \in \mathbb{R}^m \quad \hat{y} = u^T h + c$$

$$\mathcal{L} = \frac{1}{2} (u^T \cdot \sigma(w x + b) + c - y)^2$$

$$\frac{1}{2} (\hat{y} - y)^2 = \mathcal{L}(\hat{y}, y)$$

$$\frac{\partial \mathcal{L}}{\partial u} = \sigma(w x + b) (u^T \sigma(w x + b) + c - y) \quad (a)$$

$$\frac{\partial \mathcal{L}}{\partial c} = u^T \sigma(w x + b) + c - y$$

$$\frac{\partial \mathcal{L}}{\partial w} = (u^T \sigma(w x + b) + c - y) \cdot u^T \sigma(w x + b) \cdot x \quad (b)$$

$$\frac{\partial \mathcal{L}}{\partial b} = (u^T \sigma(w x + b) + c - y) \cdot u^T \sigma(w x + b)$$

$$\text{כאשר } \hat{y} - y \text{ הוא המרחק בין } \hat{y} \text{ ל-} y \quad \text{אז}$$

$$L_i(z) = CE(e_i, z) = - \sum_{j=1}^k (e_i)_j \log(z_j) = -\log(z_i)$$

.8

$$J_z(L_i) = \left[ \frac{\partial L_i}{\partial z_1}, \dots, \frac{\partial L_i}{\partial z_k} \right]$$

$$\forall j \neq i \quad \frac{\partial L_i}{\partial z_j} = 0, \quad \frac{\partial L_i}{\partial z_i} = -\frac{1}{z_i} \Rightarrow [0, \dots, -\frac{1}{z_i}, 0, \dots, 0]$$

